Failure Path Register

Assignment 2: Logistics & Shipment Exception Management (BPMN 2.0). Companion to **shipment_exception_management.bpmn**. Name / Reg. No.: _____

1. Process scope and happy path

Starts when a shipper books a shipment. **Ends when** the shipment is delivered and closed, returned to sender, or closed as lost, cancelled or claim-settled. One pool with seven lanes: Shipper / Customer, Control Tower / Dispatcher, Pickup Agent, Hub / Warehouse, Delivery Agent, Recipient, Claims Department.

#	Step	Lane	BPMN element	Why this type
1	Shipper books a shipment	Shipper	Start event (none)	Triggered by the shipper's own action.
2	Create booking & enter shipment details	Shipper	User task	A person enters the data.
3	Validate address, weight & service level	Control Tower	Business rule task	Rule-based check (serviceable pin code, weight limits) that can change without redrawing the process.
4	Calculate charges & confirm booking	Control Tower	Service task	Fully automated pricing. Has a compensation boundary so charges can be refunded.
5	Schedule pickup & assign agent	Control Tower	Service task	Automated dispatch logic.
6	Collect parcel & scan	Pickup Agent	User task	Agent acts and scans. Timer boundary covers the pickup window.
7	Parcel picked up (fork)	Pickup Agent	Parallel gateway (split)	Two things happen together: the shipper is told and the parcel travels to the hub.
8a	Notify shipper: picked up, tracking link	Control Tower	Send task	One-way message.
8b	Transport parcel to origin hub	Pickup Agent	Manual task	Physical work with no system involved.
9	Scan parcel at origin hub & update tracking	Hub	Service task	Automated scan and tracking update. Error boundary covers system failure.
10	Join (both branches done)	Hub	Parallel gateway (join)	Waits for notification and arrival so that the process does not move on half done.
11	Special handling needed?	Hub	Inclusive gateway (split and join)	International, fragile or insured can be true in any combination, or none at all.
12	Customs clearance / Inspect parcel for damage	Hub	Sub-process; user task	Customs is a self-contained scope with its own failure handling. Inspection needs a person.
13	Line-haul transit	Hub	Sub-process with manual task	One scope on which a cancellation message and a transit error can be caught.
14	Arrival scan?	Hub	Event-based gateway	The next step depends on which event happens first: the scan arrives or 48 hours pass.
15	Scan at destination hub; Sort & hand over	Hub	Service task; user task	Automated scan, then a person hands parcels to the delivery agent.
16	Attempt delivery; Recipient present?; Parcel accepted?	Delivery Agent	User task; exclusive gateways	Agent acts. Each gateway has one outcome only.
17	Cash on delivery? / Pay cash on delivery	Delivery Agent / Recipient	Exclusive gateway; user task	Only COD shipments pay. The recipient performs the payment.
18	Capture proof of delivery	Delivery Agent	Service task	Captured automatically from the handheld device.
19	Close shipment & generate invoice	Control Tower	Service task	Automated closure and billing.
20	Shipment delivered & closed	Control Tower	End event (none)	Successful completion.

2. Gateway and event choices

Construct	Where used	Reason
Exclusive gateway	Booking valid?, attempt counters, Parcel collected?, Damage severity?, Parcel found?, Recipient present?, Parcel accepted?, Cash on delivery?, Payment successful?, Claim decision	Each is a single either/or or one-of-three decision.
Exclusive gateway (merge)	Entry to booking, hub scan merge, arrival merge, return-to-sender merge, claims intake	Several paths converge on one step. Only one token arrives at a time.
Parallel gateway	Fork at "Parcel picked up", join at the origin hub	The shipper notification and the physical movement are independent and both must finish.
Inclusive gateway	Special handling needed? and its join	International customs and damage inspection are independent conditions. Both, one or neither can apply. Default flow = standard handling.
Event-based gateway	Customs wait (documents vs 5 days); Arrival scan? (scan vs 48 h); Redelivery wait (request vs 3 days)	The path is chosen by whichever event happens first, not by data.
Timer boundary (interrupting)	Pickup window (4 h)	Cuts off the wait and starts the missed-pickup handling.
Timer boundary (non-interrupting)	ETA exceeded (24 h) on transport	Transport continues while the delay notice is sent.
Error boundary / error end	Scanning system down; Breakdown / capacity; Customs hold; Transit failure	A technical or operational failure inside a scope is thrown and caught by the parent.
Message boundary	Shipper cancels after dispatch (on Line-haul transit)	A cancellation can arrive at any moment during transit and must interrupt it.
Compensation	Calculate charges (boundary), Refund charges less cancellation fee, Undo booking charges (throw)	Charges already taken must be reversed when pickup fails.
Script task	Count correction attempt, Count pickup attempt, Count delivery attempt, Count evidence request	Counters that make every loop finite.

3. Failure Path Register

ID	Failure point	Cause / trigger	Where in the model	Solution designed in the diagram	BPMN constructs	End state / outcome
F1	Invalid booking	Unserviceable address, overweight, wrong service level.	Validate address, weight & service level; "Booking valid?" (Invalid).	Count correction attempt. Under 2: booking returned with an error list and the shipper corrects it. At 2: rejected.	Business rule task, exclusive gateways, script counter, send task, loop.	Back to booking, or Closed: booking rejected.
F2	Pickup fails	Agent no-show, shipper absent, wrong or incomplete address.	Boundary timer on Collect parcel & scan; "Parcel collected?" (Not collected).	Count pickup attempt. Under 3: reschedule and assign again (address corrected). At 3: cancel booking, bill a cancellation fee, undo the charges.	Interrupting timer, script counter, loop to Schedule pickup, compensation throw event and handler.	Rescheduled, or Closed: cancelled (pickup failed).
F3	Scanning or tracking system down	Hub scanner or tracking service outage.	Error boundary on Scan parcel at origin hub.	Hub staff log the scan manually and it syncs later, so the parcel is never stuck.	Error boundary event, user task, merge gateway.	Process continues.
F4	Customs documents incomplete	Missing invoice, HS code or ID on an international shipment.	Customs clearance: "Documents complete?" (No).	Request documents from the shipper, then wait. Documents received: re-validate. 5 days with no reply: customs hold.	Send task, event-based gateway, message catch, timer catch, loop, error end.	Re-validated, or customs hold (F5).
F5	Customs hold unresolved	Documents never supplied, or clearance refused.	Error end "Customs hold unresolved"; error boundary on Customs clearance.	Error caught by the sub-process boundary and routed to return to sender.	Error end event, error boundary event, merge into return-to-sender.	Closed: returned to sender.
F6	Damage at the origin hub	Crushed, leaking or opened parcel found at inspection.	Inspect parcel for damage; "Damage severity?".	None or minor: repack and annotate condition, then continue. Severe: open a claim.	Inclusive branch, user task, exclusive gateway, flow to claims.	Continues, or claim handling (F15).
F7	Vehicle breakdown or capacity shortfall	Truck fault, route closed, no cargo space.	Error boundary on Transport to destination hub.	Reassign vehicle or re-route, then try again. Counted: fewer than 3 reassignments allowed.	Error boundary, user task, exclusive gateway with limit, loop.	Transport resumes, or F8.
F8	Transit failure	3 reassignments used without success.	Error end "Transit failure"; error boundary on Line-haul transit.	The parcel is traced across hubs.	Error end event, error boundary event, user task.	Found (continues) or lost (F12).
F9	Delay (weather, congestion)	ETA exceeded by 24 hours.	Non-interrupting timer on Transport to destination hub.	Notify the recipient of the delay and the new ETA while transport continues.	Non-interrupting timer boundary, send task, end event.	Delay notified; transport unaffected.
F10	Cancellation after dispatch	Shipper cancels while the parcel is in transit.	Message boundary on Line-haul transit.	Transit is interrupted. Control Tower stops the shipment and holds the parcel, then arranges return and bills the return fee.	Message boundary (interrupting), user task, merge gateway.	Closed: returned to sender.
F11	Missing arrival scan	Parcel not scanned at the destination hub.	Event-based gateway "Arrival scan?" (48 h timer).	If no scan in 48 hours, trace the parcel instead of waiting.	Event-based gateway, message catch, timer catch, user task.	Trace (F12) or normal flow.
F12	Parcel lost	Trace across all hubs does not find the parcel.	"Parcel found?" (No: declare lost).	Found: rejoin normal flow. Not found: open a claim.	Exclusive gateway, flow to claims.	Claim handling (F15).
F13	Recipient unavailable	Nobody at the address, wrong address, locked premises.	"Recipient present?" (No).	Count delivery attempt. Under 3: leave a notice and message the recipient, then wait for a redelivery request or 3 days. At 3 attempts or 3 days: return.	Script counter, send task, event-based gateway, message catch, timer catch, loop to Attempt delivery.	Redelivery, or return to sender.
F14	Recipient refuses the parcel	Order cancelled, wrong item.	"Parcel accepted?" (Refused).	Parcel stays with the agent and goes to return to sender.	Exclusive gateway, merge into return-to-sender.	Closed: returned to sender.
F15	Damage reported on receipt	Recipient finds damage at the door.	"Parcel accepted?" (Damaged); severe hub damage (F6); lost parcel (F12).	Claims Department opens a claim, collects evidence and decides. Approved: pay compensation. Rejected: close. More evidence: ask again, max 2 requests.	Exclusive gateways, user tasks, script counter, loop, service task.	Closed: claim settled, or claim rejected.
F16	Cash-on-delivery payment fails	Recipient cannot or will not pay.	"Payment successful?" (Payment failed).	The parcel is not handed over and goes to return to sender.	Exclusive gateways, user task, merge.	Closed: returned to sender.

4. Loop limits and end states

Loop	Limit	What happens at the limit
Booking correction (F1)	2 attempts	Closed: booking rejected
Pickup attempts (F2)	3 attempts	Cancel booking, bill fee, undo charges
Transit reassignments (F7)	3 reassignments	Transit failure, trace parcel
Delivery attempts (F13)	3 attempts or 3 days	Return to sender
Claim evidence requests (F15)	2 requests	Closed: claim rejected (no evidence)
Customs document wait (F4)	5 days	Customs hold, return to sender
Missing arrival scan (F11)	48 hours	Trace parcel

End events. Success: Shipment delivered & closed. Closed: booking rejected, cancelled (pickup failed), returned to sender, claim settled (damaged / lost), claim rejected, claim rejected (no evidence). Side ends: Delay notified. Error ends caught by a boundary: Customs hold unresolved, Transit failure.